

EE1060: Differential Equations and Transform Techniques**Practice Problem Set - 04**

Practice problems in topics covered till: Lecture 12

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

1. Consider the ODE

$$\frac{dy}{dt} = y^2 - 4y + 3 \quad (1)$$

- (a) Find the equilibrium points of the system.
- (b) Determine if the equilibrium points are locally stable.
- (c) Find the linearized differential equation around each equilibrium point

2. Consider the ODE

$$\frac{dy}{dt} = y(2 - y + y^2) \quad (2)$$

- (a) Find the equilibrium point/s of the system.
- (b) Determine if the equilibrium point/s is/are locally stable.
- (c) Find the linearized differential equation around each equilibrium point

3. Consider the ODE

$$\frac{dy}{dt} = \mu y + y^3 \quad (3)$$

- (a) Plot the bifurcation diagram for the system.
- (b) For $\mu = 1$, draw the phase plane trajectories.

4. Consider the ODE [1]

$$\frac{dy}{dt} = 1 + \mu y + y^2 \quad (4)$$

- (a) Plot the bifurcation diagram for the system.
- (b) For $\mu = 2$, draw the phase plane trajectories.

5. Consider the ODE [1]

$$\frac{dy}{dt} = \mu^2 - y^2 \quad (5)$$

- (a) Plot the bifurcation diagram for the system.

(b) For $\mu = 2$, draw the phase plane trajectories.

6. Consider the ODE [1]

$$\frac{dy}{dt} = y(\mu - e^y) \quad (6)$$

(a) Plot the bifurcation diagram for the system.

(b) For $\mu = 2$, draw the phase plane trajectories.

7. Consider the ODE [1]

$$\frac{dy}{dt} = \mu - 3y^2 \quad (7)$$

(a) Plot the bifurcation diagram for the system.

(b) For $\mu = 2$, draw the phase plane trajectories.

8. Consider the ODE [1]

$$\frac{dy}{dt} = \mu y + 4y^3 \quad (8)$$

(a) Plot the bifurcation diagram for the system.

(b) For $\mu = 2$, draw the phase plane trajectories.

9. Plot the bifurcation diagram for the ODE [1]

$$\frac{dy}{dt} = \mu y + y^3 - y^5 \quad (9)$$

Additionally, draw the phase plane trajectories for $\mu = 1$. Analyze the types of bifurcations and the corresponding bifurcation points for the system.

10. Compute the Wronskian of the following set of functions

(a) $y_1(x) = e^x$ and $y_2(x) = \sin x$.

(b) $y_1(x) = x^n$ and $y_2(x) = x^m$ with $m \neq n$.

(c) $y_1(x) = \cos mx$ and $y_2(x) = \cos nx$ with $m \neq n$.

(d) $y_1(x) = x$ and $y_2(x) = \ln x$.

(e) $y_1(x) = x^2 e^x$ and $y_2(x) = x^3$.

Please Note: Some of the problems presented in this document have been adapted from textbooks referenced for this course. These problems are included to help students practice and understand key concepts. The material is used for educational purposes only. We have made every effort to provide proper citations, but if any are inadvertently omitted, it was not intentional. All copyright and trademark rights belong to the original authors and publishers.

References

- [1] S. H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd ed. Cambridge, MA: Perseus Books, 2018.